Data Models, Schemas, and Storage
The Structural Foundations of Modern Time-Series and Panel Analytics

- Insight, prediction, and inference all depend on the *quality, structure, and governance* of the underlying data.
- Without reliable data, even the best models collapse into noise.

Data Models, Schemas, and Storage: The Backbone of Data Science

- Rigorous empirical work depends on coherent, well-governed data.
- A unified platform supports reproducibility, lineage, scalability, and multi-modal integration.
- 70–80% of analytical effort concerns locating, reconciling, and shaping data.

Data - Beyond Just Tables

Architecture defines the rules of engagement

- Data model – how real-world concepts (entities, relationships, events) are represented.
- Schema – the structure: fields, types, constraints, keys, relationships.
- Metadata – lineage, definitions, timestamps, provenance, quality.

Data comes in many shapes

- **Structured:**
  - ✓ Relational tables, time series, panel data (multi-entity, multi-period).
- **Semi-structured**:
  - ✓ JSON, XML, logs, API payloads, events.
- **Unstructured**:
  - ✓ Text, documents, media, transcripts.
- **Graph data**:
  - ✓ Entities + relationships, knowledge graphs, temporal networks.
- **Event streams**:
  - ✓ Clicks, trades, sensor signals, real-time updates.

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Data - Beyond Just Tables

Different data requires different persistence paradigms

- **Relational** (OLTP, OLAP): strong schema, joins, integrity.
- **Document stores**: flexible JSON objects, evolving schemas.
- **Columnar stores**: analytics-optimized, high compression, fast scanning.
- **Time-series databases**: retention, fast append, window ops.
- **Graph databases**: relationships as first-class citizens.

Schema - The Blueprint Behind Your Data

A data schema - *blueprint* that describes how data is organized.

- **Fields / Columns** (e.g., date, price, country)
- **Data types** (numeric, categorical, boolean, timestamp, JSON, etc.)
- **Relationships** (
  - ✓ Keys (primary, foreign)
  - ✓ Parent–child hierarchies
  - ✓ One-to-one, one-to-many, many-to-many
  - ✓ Temporal links (events tied to entities)
  - ✓ Graph edges (entity–entity relationships)
- **Constraints** (valid ranges, uniqueness, null rules)
- **Metadata** (definitions, units, lineage)

Schema-on-Write: Structure First;

Data is **validated, typed, and shaped** at ingestion.

Schema-on-Read: Flexibility First

Schema - The Blueprint Behind Your Data

Schema-on-Write: Structure First

- Data is validated, typed, and shaped at ingestion.
- Guarantees governance, consistency, referential integrity, and data quality.
- Essential for transactional and highly structured environments:
  - ✓ Operational databases
  - ✓ Enterprise data warehouses
- Optimized for fast updates, indexing, joins, and highly structured queries

Schema-on-Read: Flexibility First

- Raw data lands as-is; no predefined schema imposed at ingestion.
- Structure is assigned only at query time, depending on the analysis
- Ideal for semi-structured and unstructured sources:
  - ✓ JSON, events, logs
  - ✓ Text, audio, PDFs, images
  - ✓ Web scrapes, API dumps
- Supports exploration, NLP, anomaly detection, ML feature extraction.

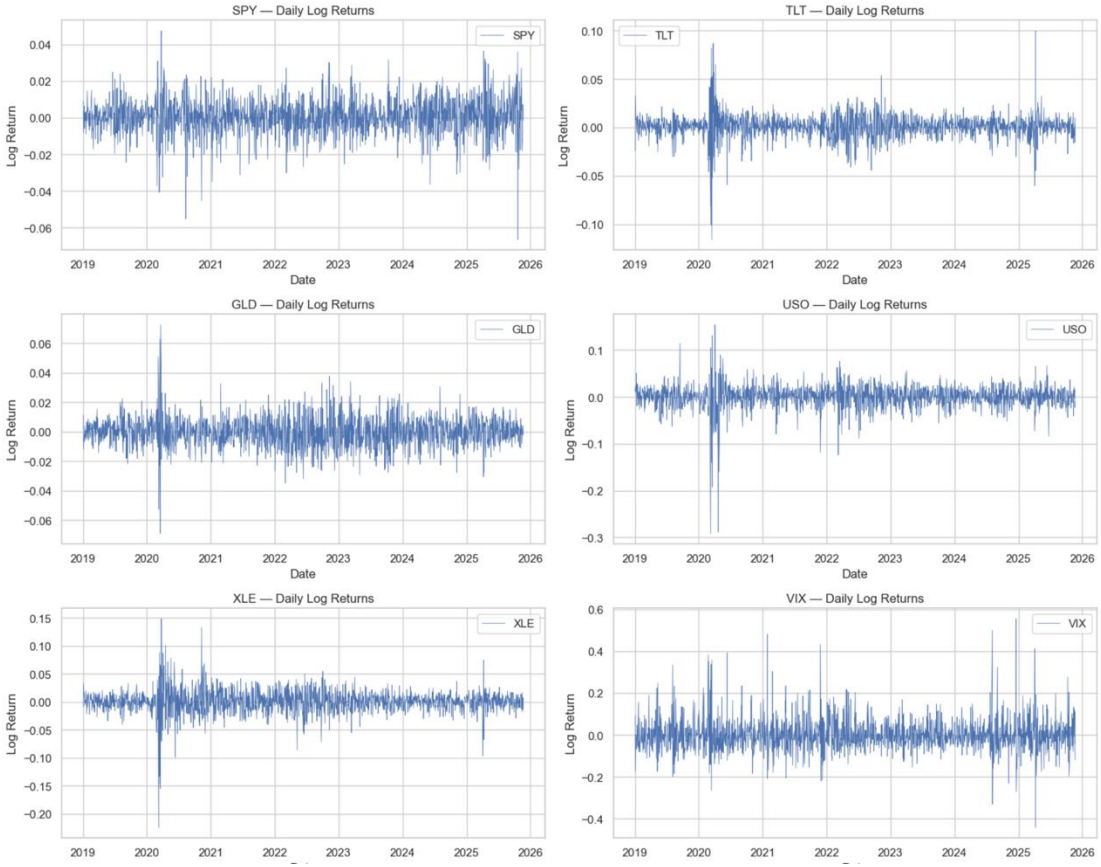
Data warehouses: denormalized, columnar, still schema-on-write

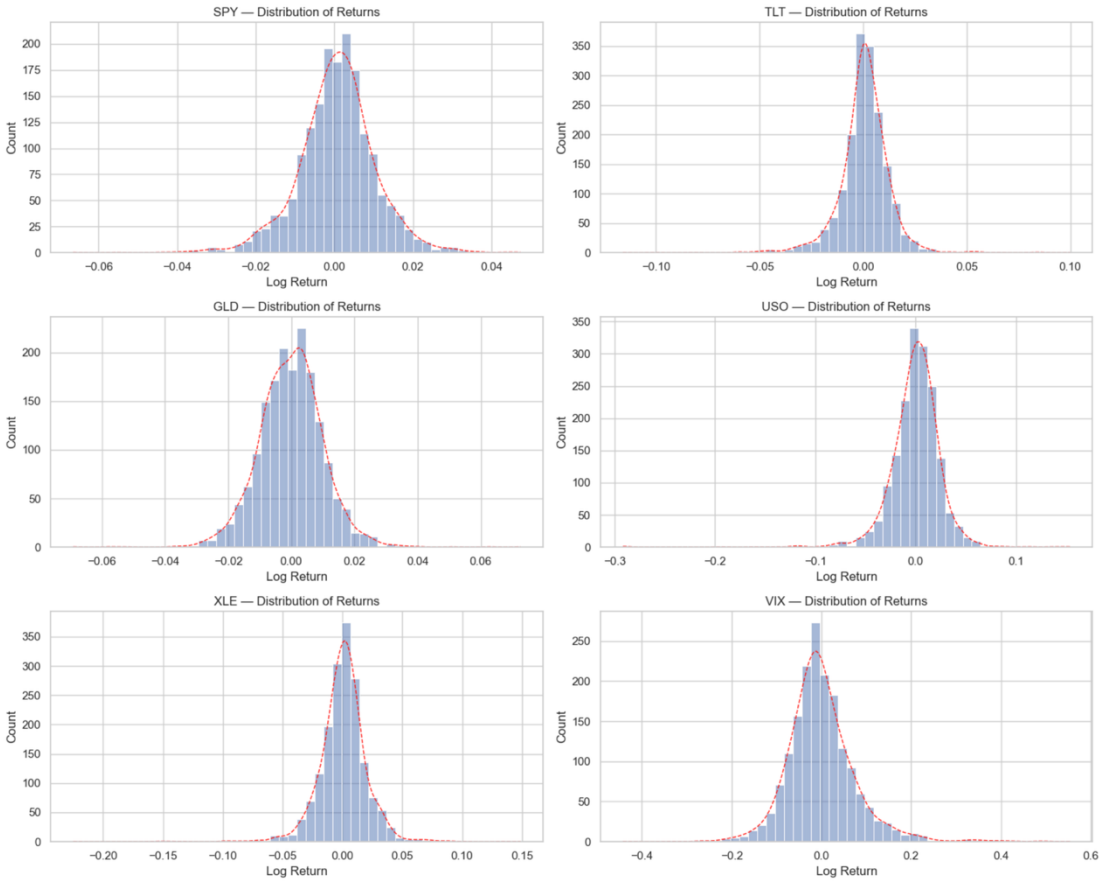
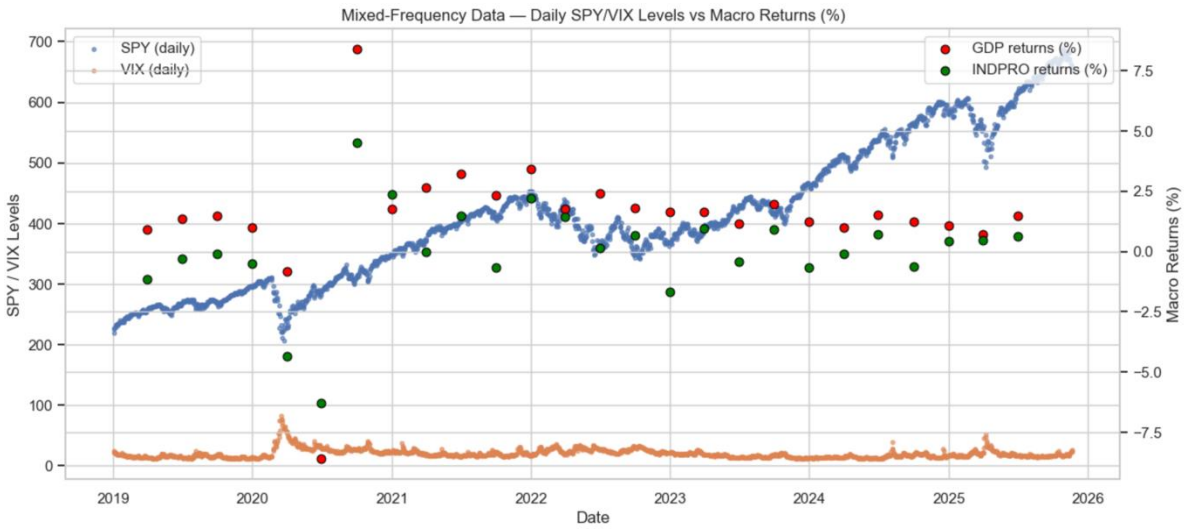
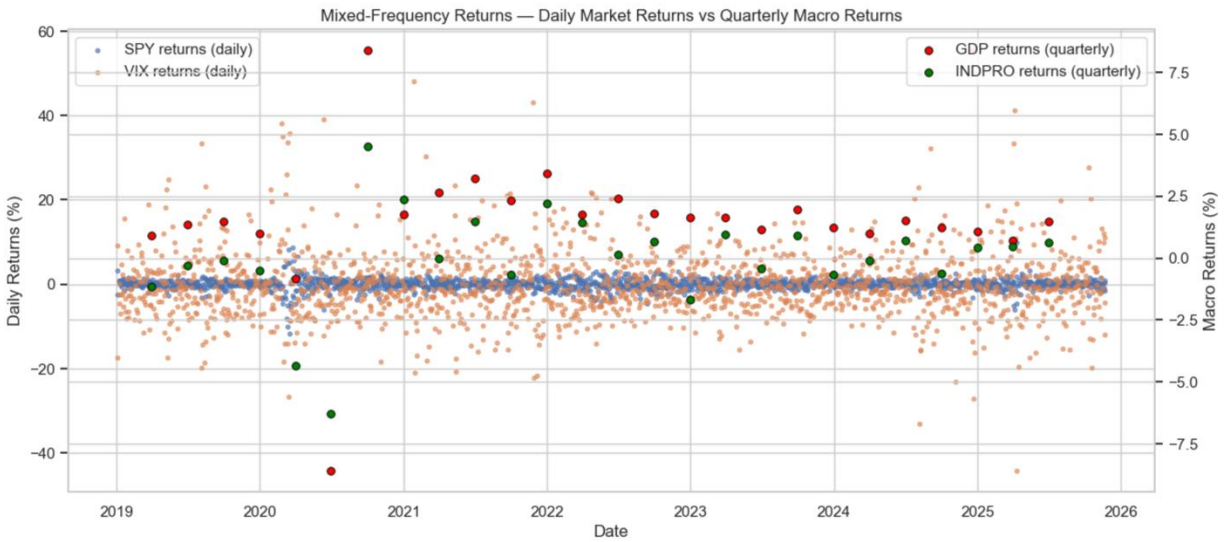
- For analytics: high-speed scanning, aggregations

Data lakes / lakehouses: raw object storage + curated structured layers

- Hybrid: schema-on-read *and* schema-on-write together
- Handle text, events, time series, logs, graph data, and structured tables in one place

Time-Series Data - Univariate





Time-Series Data – Multivariate, Different Frequency